
When Redundant Probes Help in Strategic Auditing

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Abstract

Audits often favor diverse probes to cover a broad range of failures. We study when this choice helps against a strategic agent that knows the audit policy and chooses an exploit to maximize expected payoff. In a geometric model, we compare two randomized audits: uniform sampling of probes, and volume-weighted sampling, which draws probe subsets with probability proportional to the volume they span, so near-duplicate probes are rarely selected together (a determinantal point process). When probes cluster around a dominant subspace, volume weighting improves protection against exploits in that subspace. Uniform sampling can perform better against exploits in weakly measured residual directions: differences between near-duplicate probes provide coverage that volume weighting tends to suppress. We derive the limiting average residual coverage and observe a crossover between policies in synthetic experiments. The uniform policy’s payoff advantage is small despite a larger coverage ratio, and declines as regularization attenuates weak probe differences. These results identify a trade-off between diversity and residual coverage and motivate an audit-selection heuristic based on exploit geometry.

1 Introduction

Diversity is a common objective in robustness evaluation, including quality-diversity search for failures, red-teaming coverage, and the selection of complementary evaluation probes. A diverse audit may cover many behaviors, but a strategic agent can exploit directions that remain poorly monitored. This concern arises in reward hacking under finite evaluation [Wang and Huang, 2026, Skalse et al., 2022], tool-using agents that exploit verifier weaknesses [Thaman, 2026], and budgeted monitoring for AI control [Greenblatt et al., 2024].

Randomization is a standard response to predictable inspection schedules [Avenhaus et al., 2002, Blocki et al., 2013, Tambe, 2011] and can also help in strategic classification [Braverman and Garg, 2020]. We ask a more specific question: *among randomized audits, when does selecting more diverse probes improve protection?* We compare uniform sampling with volume-weighted sampling, a standard approach to selecting complementary features [Belhadji et al., 2020].

Our model represents behavior as a vector and each probe as a measurement direction. Probes form clusters around a *dominant subspace*; its orthogonal complement is the *residual subspace*. Small perturbations give individual probes weak sensitivity to the residual. Selecting near-duplicate probes can nevertheless help: their differences introduce additional audited directions. Uniform sampling retains such pairs more often than volume-weighted sampling (Fig. 1). This mechanism concerns weakly measured directions, not directions absent from the full probe span.

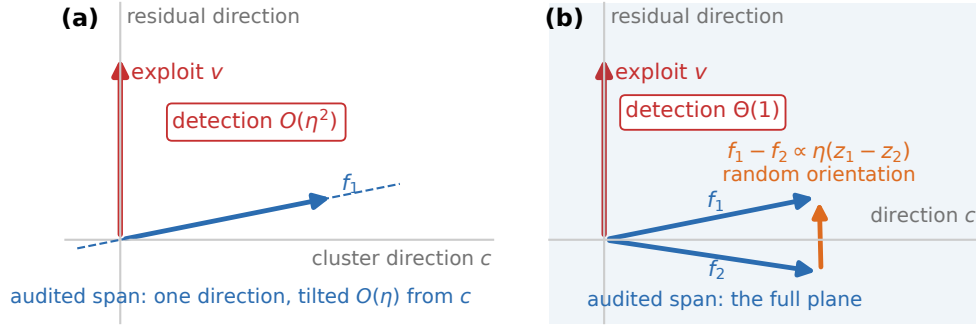


Figure 1: How near-duplicate probes reveal a residual direction. (a) One probe from a cluster has only a small component along the residual, giving detection probability $O(\eta^2)$. (b) Two probes from that cluster span an additional direction, allowing detection of order one in the noiseless model. The panels illustrate the local mechanism, not a comparison at equal audit budgets.

We identify a crossover as exploitable gain shifts between subspaces (§3.2), derive the limiting residual coverage from duplicate selections (§3.3), and examine its sensitivity to regularization (§3.4). We also propose an audit-selection heuristic (§4) that remains to be validated on held-out exploits.

2 A geometric model of auditing

Let behavior lie in $\mathbb{R}^{n_a}$. The monitor has d probes formed by normalizing $c_{[i]} + \eta z_i$, where $c_{[i]}$ is the center of probe i 's cluster and $z_i \sim \mathcal{N}(0, I)$ independently. The r linearly independent centers span U ; smaller η produces more redundant probes. Write $F = [f_1, \dots, f_d]$ and $K = F^\top F$. An audit selects k probes, indexed by $S \subset [d]$, and P_S projects onto their span. We assume that a unit exploit direction v is detected with probability $\|P_S v\|^2$.

The agent knows the audit distribution but not the sampled subset. Its expected detection probability is $v^\top M v$, where $M = \mathbb{E}_S[P_S]$. For a positive semidefinite gain matrix G , the agent's best-response payoff is

$$u^*(M) = \max_{\|v\|=1} (v^\top G v)(1 - v^\top M v), \quad G_\alpha = (1 - \alpha)Q_U Q_U^\top + \alpha Q_\perp Q_\perp^\top. \quad (1)$$

Here Q_U and $Q_\perp$ are orthonormal bases of U and $U^\perp$. In the subspace experiments, $\alpha \in [0, 1]$ controls gain per direction in the residual relative to the dominant subspace. Lower payoff means better protection.

We compare two deterministic policies, top- k selection by probe norm and greedy maximum-volume selection, with uniform sampling and a k -DPP [Kulesza and Taskar, 2012]. The latter assigns probability proportional to $\det(K_{S,S})$, favoring subsets with larger squared volume. We compute best responses with multi-start L-BFGS-B; Appendix A describes implementation details and numerical checks. Proposition 1 states the conditions under which randomization improves the worst-case payoff.

3 Results

3.1 Comparing deterministic and randomized audits

The main experiment uses $n_a = 25$, $d = 40$, $r = 8$, $\eta = 0.25$, 12 seeds, and budgets $k \in \{2, 4, 6, 8, 10, 12\}$. Probe norms vary slightly, and G has a random rank-five component plus an isotropic term (Appendix A). The k -DPP dominates each deterministic policy in 10 of 12 seeds and is never dominated by either (Fig. 2, left). Dominance means no higher payoff at any budget and lower payoff at least once. Uniform sampling and the DPP show no consistent ordering (Appendix B).

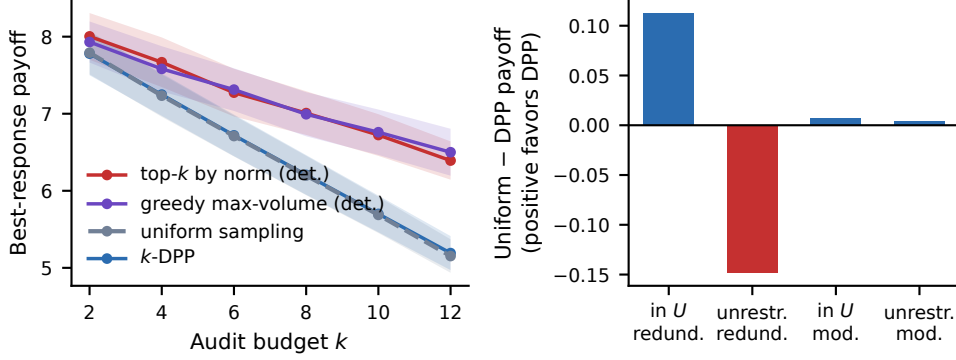


Figure 2: Comparison of audit policies. Left: mean best-response payoff across 12 seeds; shaded bands show one standard error. Right: mean uniform-minus-DPP payoff gap across eight seeds per setting for redundant (redund., $\eta = 0.05$) and moderately redundant (mod., $\eta = 0.25$) probes, with gain confined to U or unrestricted (random rank five, mostly outside U). Positive values favor the DPP. With strongly redundant probes, the preferred policy depends on where the gain lies.

3.2 A crossover as exploit gain shifts

With strongly redundant probes ($\eta = 0.05$), the DPP dominates uniform sampling in 6 of 8 seeds when gain is confined to U ; uniform sampling dominates the DPP in all 8 seeds when gain is unrestricted and therefore lies mostly outside U (Fig. 2, right). Sweeping α reveals a crossover near $\alpha^* = 0.49$ for $\eta \in \{0.02, 0.05, 0.1\}$ (Fig. 3, left). At $\alpha = 0$, the uniform-minus-DPP gaps are 0.084, 0.058, and 0.020, respectively. At $\alpha = 1$, they are -0.0035 , -0.0029 , and -0.0018 , with paired bootstrap 95% confidence intervals excluding zero at all three η (per-seed values in the code archive). The difference between policies becomes small for $\eta \gtrsim 0.2$.

The uniform policy’s residual advantage is substantially smaller than the DPP’s dominant-subspace advantage. A two-subspace approximation predicts a crossover at 0.41–0.43, below the observed 0.49 (Appendix D).

3.3 Residual coverage from duplicate selections

As $\eta \rightarrow 0$, the span of two probes from the same cluster approaches the span of their center and a perturbation-difference direction. After removing the center component, the latter’s orientation remains random even as its magnitude shrinks. Uniform sampling therefore retains residual directions that the DPP increasingly suppresses.

Lemma 1 (Limiting residual coverage). *Fix $r < n_a$ linearly independent cluster centers and let $k \leq r$. Let D be k minus the number of distinct clusters in a uniformly sampled k -subset. Averaging over Gaussian perturbations Z and audit subsets, as $\eta \rightarrow 0$, $\mathbb{E}_Z[\text{tr}(Q_\perp^\top M Q_\perp)/(n_a - r)] \rightarrow \mathbb{E}_S[D/(n_a - k + D)]$ for uniform sampling, whereas the corresponding k -DPP coverage is $\Theta(\eta^2)$.*

Appendix E gives the proof sketch. For $k = 4$, $r = 6$, $d = 40$, and $n_a = 25$, the predicted floor is 0.036, compared with 0.038 observed at $\eta = 0.02$; DPP coverage is 0.0031. Average coverage does not determine worst-case protection. The regularization experiment also measures the residual block’s minimum eigenvalue: 0.0039 for uniform sampling and 0.0004 for the DPP at $\eta = 0.02$, $\lambda = 0$ (Appendix C).

3.4 Sensitivity to regularization

The noiseless model gives full coverage to every independent selected direction, regardless of its magnitude. To test sensitivity to weak probe differences, we replace P_S with $A_{S,\lambda} = F_S(F_S^\top F_S + \lambda I)^{-1} F_S^\top$. A selected direction with singular value s then receives weight $s^2/(s^2 + \lambda)$. The

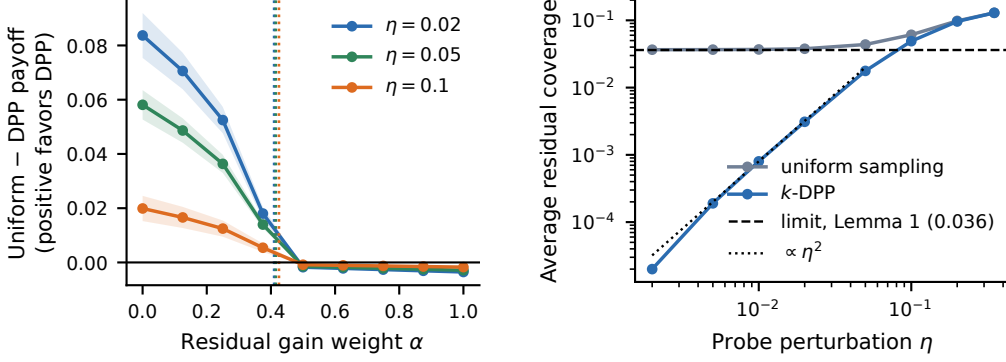


Figure 3: Policy crossover and residual coverage. Left: mean uniform-minus-DPP payoff gap across five seeds, with one-standard-error bands; dotted lines mark predicted crossovers. Right: mean residual coverage across five seeds (points below $\eta = 0.02$ from the limit check in Appendix E). The dashed line shows the limiting expectation in Lemma 1; the dotted line indicates η^2 scaling. At $\eta = 0.02$, uniform sampling gives about 12 times the DPP’s average residual coverage.

unnormalized difference has expected squared norm $2n_a\eta^2$, suggesting a scale for attenuation, rather than an exact noise threshold.

At $\eta = 0.02$, increasing λ through $0, 10^{-3}, 10^{-2}, 10^{-1}$ reduces uniform residual coverage from 0.038 to 0.034, 0.018, and 0.004. Its payoff advantage at $\alpha = 1$ falls from 0.0035 to 0.0031, 0.0015, and 0.0002, while the DPP’s advantage at $\alpha = 0$ remains between 0.069 and 0.078 (Appendix C). The benefit of redundant probes thus depends on the audit’s sensitivity to small differences.

4 A heuristic for choosing an audit policy

Exploit geometry suggests a possible selection rule. For labeled exploit vectors $g_1, \dots, g_m$, define the dimension-normalized energy statistic

$$\hat{\alpha} = \frac{E_{\perp}/(n_a - r)}{E_U/r + E_{\perp}/(n_a - r)}, \quad E_U = \sum_j \|Q_U^{\top} g_j\|^2, \quad E_{\perp} = \sum_j \|Q_{\perp}^{\top} g_j\|^2.$$

If the vectors have second moment proportional to G_{α} , this statistic converges to α under standard sampling assumptions. Dimension normalization matters: the raw residual energy fraction is 0.76 at $\alpha = 0.5$ when $n_a = 25$ and $r = 6$.

One could estimate each policy’s coverage matrix and use the predicted crossover to choose the DPP when $\hat{\alpha} < \alpha^*$, and uniform sampling otherwise. This heuristic has not been evaluated on held-out exploits. In particular, samples of caught exploits may reflect the current audit’s coverage and need not satisfy the assumed second-moment model.

5 Discussion and limitations

Our experiments use Gaussian probe clusters, a single exploit direction, and one audit decision. They do not isolate the effect of strategic adaptation from that of a fixed exploit distribution. The two-subspace approximation omits cross-block terms of M , whose contribution to its roughly 0.07 crossover error remains untested. Regularization measures sensitivity to weak directions but does not define a calibrated noisy detector. Actual monitor responses on agent traces [Thaman, 2026] and sequential audits are needed to assess the practical value of the mechanism.

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A Implementation and solver checks

In the main experiment, each normalized probe is multiplied by an independent factor sampled uniformly from $[0.8, 1.2]$. The top- k baseline ranks squared probe norms, K_{ii} ; these scores are not statistical leverage scores. The gain matrix is $G = BB^\top/5 + 0.05I$, where $B \in \mathbb{R}^{25 \times 5}$ has independent standard Gaussian entries. In the subspace and regularization experiments, probes have unit norm and the implementation uses $G_\alpha + 10^{-3}I$. Cluster membership is assigned cyclically over the r centers.

The solver maximizes Eq. (1) using L-BFGS-B on the normalized vector $v = x/\|x\|$, with an analytic gradient. The main experiment uses 80 random initializations; the subspace sweep uses 50 and the regularization sweep uses 40. Additional initializations use the eight leading eigenvectors of G and $(I - M)^{1/2}G(I - M)^{1/2}$, together with normalized nonzero vectors $(I - M)q$ for those eigenvectors q of G .

At $M = 0$, the solver recovers $\lambda_{\max}(G)$. On a deterministic-projector instance, it matches a 300-start local search at a payoff of 6.983; an earlier Riemannian ascent implementation returned 5.585 on that instance, and its results were discarded. Every reported payoff also satisfies the upper bound

$$u^*(M) \leq \lambda_{\max}(G)(1 - \lambda_{\min}(M)).$$

This bound is a consistency check; satisfying it does not certify global optimality. Coverage matrices are estimated using 1,500 sampled subsets per randomized policy in the main experiment, 800 in the subspace sweep, and 600 in the regularization sweep. For the smallest reported effect (the $\alpha = 1$ gap at $\eta = 0.02$), quadrupling the sampled subsets and tripling the optimizer starts changes the gap by at most 0.0003 in each of three seeds, against a gap of 0.0035.

B Additional results

Table 1 compares policies across all tested budgets. A policy dominates another if it has no higher best-response payoff at any budget and a lower payoff at least once. Table 2 reports average residual coverage in the perturbation sweep.

Table 1: Policy comparisons across 12 seeds and budgets $k \in \{2, 4, 6, 8, 10, 12\}$. Each row compares the k -DPP with one baseline.

Baseline	DPP dominates	Baseline dominates	Neither dominates
Top- k by norm	10	0	2
Greedy volume	10	0	2
Uniform	2	2	8

Table 2: Average residual coverage $\text{tr}(Q_{\perp}^{\top} M Q_{\perp}) / (n_a - r)$ across five seeds, with $k = 4$.

η	0.02	0.05	0.10	0.20	0.35
Uniform	0.0378	0.0437	0.0610	0.0981	0.1294
k -DPP	0.0031	0.0177	0.0490	0.0962	0.1293

The ablation experiment uses $n_a = 25$, $d = 40$, $r = 6$, budgets $k \in \{2, 3, 4, 6\}$, and eight seeds per setting. Gain is either confined to U (a random positive semidefinite form on the cluster centers) or unrestricted (a random rank-five matrix over $\mathbb{R}^{n_a}$ plus $0.05I$, most of whose mass lies outside U). For strongly redundant probes ($\eta = 0.05$), the DPP dominates uniform sampling in six seeds with gain confined to U , whereas uniform sampling dominates in all eight seeds with unrestricted gain. At $\eta = 0.25$, neither policy dominates the other in any seed under either gain setting.

C Regularization results

Table 3 reports average and minimum residual coverage, together with payoff gaps at the two endpoints of the gain sweep. Each entry is averaged over five seeds. The seeds and Monte Carlo sample count differ from those in Table 2, so the unregularized estimates need not coincide.

Table 3: Regularized audits with $k = 4$. C is average residual coverage and μ is the residual block’s minimum eigenvalue. U denotes uniform sampling and D denotes the DPP. $\Delta_{\alpha} = u_{\text{U}}^* - u_{\text{D}}^*$; positive values favor the DPP.

η	λ	C_{U}	C_{D}	μ_{U}	μ_{D}	Δ_0	Δ_1
0.02	0	.0382	.0034	.0039	.0004	+.069	−.0035
0.02	10^{-3}	.0342	.0033	.0035	.0004	+.071	−.0031
0.02	10^{-2}	.0183	.0026	.0019	.0003	+.078	−.0015
0.02	10^{-1}	.0042	.0018	.0005	.0002	+.069	−.0002
0.05	0	.0444	.0181	.0054	.0023	+.036	−.0030
0.05	10^{-2}	.0380	.0167	.0046	.0021	+.038	−.0024
0.05	10^{-1}	.0193	.0118	.0024	.0015	+.037	−.0009
0.10	0	.0617	.0503	.0085	.0067	+.015	−.0017
0.10	10^{-1}	.0455	.0398	.0061	.0052	+.015	−.0009

D Randomization and the crossover approximation

Proposition 1 (A benefit of randomization). *Let $G \succeq 0$ with $\lambda_{\max}(G) \leq 1$. For a deterministic audit $M = P_S$ of rank at most $k < n_a$, a unit direction in $\ker P_S$ attains payoff $\lambda_{\max}(G|_{\ker P_S})$ with zero detection. Hence $u^*(M) \geq \lambda_{\max}(G|_{\ker P_S})$; the bound is positive if $\text{range}(G) \not\subseteq \text{range}(P_S)$. Now restrict unit exploit directions to a fixed subspace W of dimension $r' > k$ and consider gain matrices supported on W . If a randomized audit has $\mu_W = \min_{v \in W, \|v\|=1} v^\top M v > 0$, its worst-case payoff is at most $1 - \mu_W < 1$, whereas every deterministic audit of rank at most k has worst-case payoff 1.*

The second statement concerns the game restricted to W . Restricting only $\text{range}(G)$ does not, by itself, restrict the maximization in Eq. (1) to W .

For the crossover approximation, define $\mu_U = \lambda_{\min}(Q_U^\top M Q_U)$ and $\mu_\perp = \lambda_{\min}(Q_\perp^\top M Q_\perp)$. These scalars are the minimum coverage within the two subspaces. Ignoring cross-block terms of M gives

$$\tilde{u}(M, \alpha) = \max_{c \in [0,1]} ((1 - \alpha)c + \alpha(1 - c))(1 - \mu_U c - \mu_\perp(1 - c)),$$

where c is the squared norm of the exploit's component in U . This expression is exact for the block-diagonal surrogate with gain G_α . In the tested redundant-probe settings, the DPP has larger μ_U and smaller $\mu_\perp$ than uniform sampling. The surrogate predicts a crossover at $\alpha^* = 0.41\text{--}0.43$, compared with the observed 0.49. Omitted cross-block terms may contribute to the discrepancy, but that explanation has not been tested.

E Proof sketches

For Proposition 1, choose a unit vector maximizing $v^\top G v$ over $\ker P_S$. Its detection probability is zero, so its payoff equals $\lambda_{\max}(G|_{\ker P_S})$. In the game restricted to W , a deterministic audit of rank at most $k < r'$ leaves a unit direction $v \in W \cap \ker P_S$. Taking $G = vv^\top$ gives payoff 1, which is also the largest possible payoff under the gain normalization. For a randomized audit with $\mu_W > 0$, every admissible unit $v \in W$ satisfies $(v^\top G v)(1 - v^\top M v) \leq 1 - \mu_W$. Positive coverage on W , rather than full support of the sampling distribution alone, is the condition needed for this improvement.

For Lemma 1, condition on the selected clusters and write $q = k - D$. As $\eta \rightarrow 0$, the audited subspace approaches the span of the q selected centers together with a D -dimensional subspace generated by perturbation differences. After projection orthogonal to the selected centers, that random subspace is isotropic in an $(n_a - q)$ -dimensional space containing $U^\perp$. Its expected residual trace is therefore $D(n_a - r)/(n_a - q)$. Dividing by $n_a - r$ and averaging over subsets gives $\mathbb{E}_S[D/(n_a - k + D)]$. We state only the limit: normalization and orthogonalization against perturbed rather than exact centers move directions by $O(\eta)$, and we have not shown that the resulting energy correction is $O(\eta^2)$. Numerically, uniform residual coverage decreases monotonically toward the limit, 0.0378, 0.0369, 0.0367, 0.0366 at $\eta = 0.02, 0.01, 0.005, 0.002$ (five seeds, spread ± 0.001), against the predicted 0.0362, while DPP coverage divided by η^2 stays near 8 for $\eta \leq 0.01$.

For the DPP with $k \leq r$, the limiting probe matrix has rank r and subsets with a repeated cluster have zero volume. At small positive η , the probability of selecting a repeated cluster is $O(\eta^2)$. Such selections contribute at the same order as the residual tilt of probes from distinct clusters. Both terms must be included when accounting for the DPP's $\Theta(\eta^2)$ residual coverage. The limiting uniform expectation is computed by counting subsets without replacement using the cluster sizes.